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High-power multi-wavelength distributed feedback laser array with ultra-high single-mode stability, low noise, and narrow linewidth

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We report a high-power, low-noise C-band 20-channel distributed feedback (DFB) laser array. By employing an equivalent asymmetric corrugation-pitch-modulated grating and a lightly *n*-doped passive waveguide, we effectively mitigate spatial hole burning and reduce absorption losses. The array exhibits uniform 200 GHz spacing with ±0.14 nm deviation. Packaged modules deliver fiber-coupled power exceeding 190 mW at 25°C and sustain 150 mW at 75°C. Featuring a narrow 29 kHz Lorentzian linewidth and relative intensity noise (RIN) below −160 dB/Hz, the device is ideal for optical input/output (I/O), free-space communications, and LiDAR systems.

Keywords: high power; distributed feedback laser array; linewidth; optical input/output.

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1. Introduction

The exponential proliferation of data-centric paradigms, including cloud computing, generative artificial intelligence (AI), and the Internet of Things (IoT), has imposed unprecedented demands on data center throughput[1]. As traditional electrical input/output (I/O) interfaces approach their physical bandwidth-density and power-efficiency limits, the industry is rapidly pivoting toward optical I/O technologies as a critical enabler for next-generation high-performance computing (HPC)[2]. Since the seminal demonstration of on-chip optical communication by Orcutt *et al.*[3], optical I/O solutions have garnered significant traction from major industrial players, driving the development of co-packaged optics (CPO) and chip-to-chip optical interconnects[4–7].

To satisfy the escalating bandwidth requirements of these links, dense wavelength division multiplexing (DWDM) has emerged as the architecture of choice[6–9]. Consequently, multi-wavelength laser (MWL) sources aligned with the DWDM grid have become the cornerstone of system scalability. Among various MWL solutions, distributed feedback (DFB) laser arrays are favored for their intrinsic single-mode stability and high output power[10]. Notably, commercial implementations such as the SuperNova light source have validated the efficacy of parallel DFB arrays in scaling bandwidth via $M \times N$ port distribution[6,11,12]. However, this distributed architecture introduces substantial optical power splitting losses, necessitating laser sources with significantly higher output power to close the link budget. Furthermore, scaling the channel count exacerbates fabrication challenges, particularly in maintaining high manufacturing yields while ensuring precise wavelength registration and uniform single-mode operation across the entire array.

In this paper, we design and experimentally demonstrate a high-performance 20-channel DFB laser array operating in the C-band, incorporating a suite of advanced design strategies to reconcile the trade-offs among output power, spectral purity, and fabrication yield. To ensure precise wavelength registration without the complexity of multiple holographic exposures, we leverage the reconstruction-equivalent-chirp (REC) technique. This approach is synergistically combined with an equivalent asymmetric corrugation-pitch-modulated (ACPM) grating structure, which is tailored to homogenize the longitudinal photon density profile. By effectively mitigating the longitudinal spatial hole burning (L-SHB) effect, this design secures robust single-mode stability and facilitates narrow linewidth operation. Furthermore, to maximize the output power budget required for optical I/O splitting, the device features a specially designed lightly *n*-doped passive waveguide. By guiding the optical field toward the *n*-region, this architecture significantly suppresses free-carrier absorption losses caused by the overlap between the optical mode and the *p*-type cladding as well as active layers, thereby enhancing slope efficiency. Finally, to ensure operational stability, the output waveguide is designed with a specific curvature to minimize facet back-reflections, with the grating period in the bent section carefully adjusted to compensate for feedback losses.

The fabricated array exhibits a uniform channel spacing of 200 GHz with a wavelength deviation within 0.14 nm. Under continuous-wave (CW) operation at 25°C, individual channels deliver output powers exceeding 320 mW, and the side-mode suppression ratio (SMSR) is above 65 dB. Packaged in a standard butterfly module, the device delivers a fiber-coupled output power exceeding 190 mW at a drive current of 1 A at 25°C and sustains over 150 mW even at an elevated case temperature of 75°C. Additionally, the device demonstrates a narrow intrinsic Lorentzian linewidth of 29 kHz and a relative intensity noise (RIN) floor below −160 dB/Hz, confirming its suitability for rigorous optical I/O applications.

2. Device Design and Fabrication

The improvement in slope efficiency is fundamentally achieved by minimizing internal optical losses, which are dominated by free-carrier absorption in the p-type cladding and active regions. To mitigate this, we implemented an asymmetric waveguide design incorporating a thick, lightly doped n-type InGaAsP layer. This structure effectively draws the optical mode intensity distribution toward the lower-loss n-region, thereby reducing the modal overlap with the highly absorptive p-doped layers. However, this design requires a careful trade-off: the optical mode must be shifted enough to reduce loss, but not so far as to critically reduce the optical confinement factor within the active region, which would otherwise penalize the threshold current.

In the optimized epitaxial structure, the ridge waveguide width is set to 3.2 μm. The key n-side InGaAsP waveguide layer features a thickness of 3 μm, a photoluminescence (PL) wavelength of 1.1 μm, and a low doping concentration of $5\times10^{16}\ \mathrm{cm}^{-3}$. In the proposed structure, InGaAlAs is used for the quantum wells, with the number of quantum wells being two and each having a thickness of 6 nm. The simulated optical confinement factor is 1.8%. The grating material used is InGaAsP with a PL wavelength of 1.1 μm and a thickness of 10 nm, ensuring that the coupling constant (κL) is approximately 1, thereby providing sufficient feedback. Additionally, the doping profile near the p-interface is graded to further suppress free-carrier absorption.

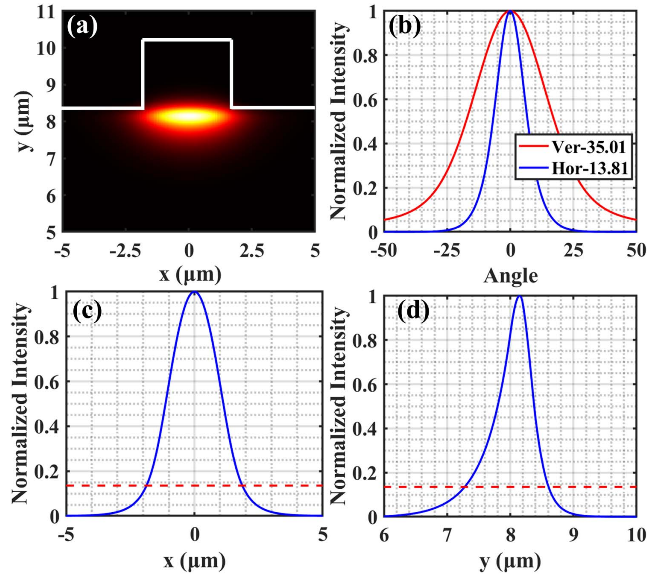

Fig. 1. Simulation results for the optimized asymmetric waveguide with a ridge width of 3.2 μm. (a) 2D mode field distribution showing the mode offset; (b) simulated far-field intensity distributions in the horizontal and vertical directions; (c), (d) simulated near-field intensity profiles in the (c) horizontal and (d) vertical directions.

Numerical simulations were performed to validate this design. Figure 1(a) illustrates the 2D optical field distribution, confirming that the mode center is effectively offset toward the n-cladding layer. The corresponding far-field characteristics are analyzed in Fig. 1(b), yielding simulated full width at half-maximum (FWHM) divergences of 13.81° (horizontal) and 35.01° (vertical). Furthermore, the near-field intensity profiles presented in Figs. 1(c) and 1(d) indicate mode field diameters (MFDs at $1/e^2$) of 3.7 and 1.3 μm in the horizontal and vertical directions, respectively. This quasi-circular mode profile is instrumental in maximizing coupling efficiency to standard single-mode fibers.

To maximize the unidirectional output efficiency, high-reflection (HR) and anti-reflection (AR) coatings are deposited on the rear and front facets, respectively. However, the random cleavage phase associated with the HR facet can severely disrupt the constructive interference condition, thereby degrading the single-mode yield of the array. In our previous investigation[13,14], we demonstrated that introducing an asymmetric phase shift into the cavity can effectively compensate for this random facet phase. Nevertheless, under high injection currents, such discrete phase shifts tend to induce a localized concentration of photon density. This localization exacerbates the L-SHB effect, which in turn deteriorates spectral stability, broadens the linewidth, and degrades noise performance[15].

To overcome these limitations, we propose an ACPM grating structure to synthesize an equivalent distributed phase shift. Unlike discrete designs, the distributed nature of the corrugation-pitch-modulated (CPM) grating homogenizes the longitudinal optical field profile, thereby significantly mitigating the L-SHB effect[16,17]. To ensure sufficient optical feedback strength while minimizing the spatial hole burning effect, the length of the ACPM structure is designed to be 1/8 of the total cavity length[16]. Furthermore, to mitigate the detrimental impact of the random cleavage phase at the HR facet on the single-mode yield, the ACPM region is positioned at a distance of 1/4 of the cavity length from the HR facet[14,17]. This specific asymmetric configuration is engineered to guarantee a high single-mode yield. Furthermore, the complex ACPM profile is implemented via the REC technique[18–22].

The fundamental principle of the REC technique relies on the Fourier analysis of sampled Bragg gratings. By precisely tailoring the sampling modulation pattern, equivalent effects such as chirp and phase shifts can be synthesized within the target high-order subgrating. This approach allows for complex phase structures to be realized without requiring nanometer-scale precision in the physical grating pitch. Usually, the +1st-order subgrating is to select the operating mode. The grating period of the +1st-order subgrating is derived as

$$\frac{1}{\Lambda_{+1}} = \frac{1}{\Lambda_0} + \frac{1}{P}, \tag{1}$$

where Λ_{+1} is the grating period of the +1st-order subgrating. The Bragg wavelength of the uniform seed grating is designed far away from the gain region to avoid lasing, while that of the +1st-order subgrating is in the center of the gain region. The lasing wavelength is determined by the sampling period. A phase-arranging section is realized by modulating the sampling period of the grating. The period of this section is expressed by

$$\frac{1}{P_P} = \frac{P_0}{1 - P_0/2L_P}, \tag{2}$$

where P_P and L_P denote the sampling period and the length of the phase-arranging section, respectively, and P_0 represents the sampling period of the regions outside this section. The inclusion of the phase-arranging section effectively induces an equivalent phase shift within the cavity. Additionally, the output waveguide is angled at 7° relative to the facet normal. This geometric design serves to minimize facet back-reflections and decouple the cavity from external optical feedback, thereby ensuring robust modal stability across a wide range of operating currents. To account for the geometric path length difference in the bent section, the sampling period of the grating is rigorously compensated. This adjustment counteracts the deviation in the effective grating pitch caused by the waveguide curvature, preserving sufficient optical feedback strength.

The design principle is further elucidated in Fig. 2(a), which depicts the schematic of the equivalent sampled grating based on the REC technique. The corresponding spectral characteristics, calculated using the transfer matrix method (TMM), are presented in Fig. 2(b). The simulation reveals a distinct, narrow transmission resonance within the stopband of the +1st-order subgrating. The presence of this sharp peak confirms that the

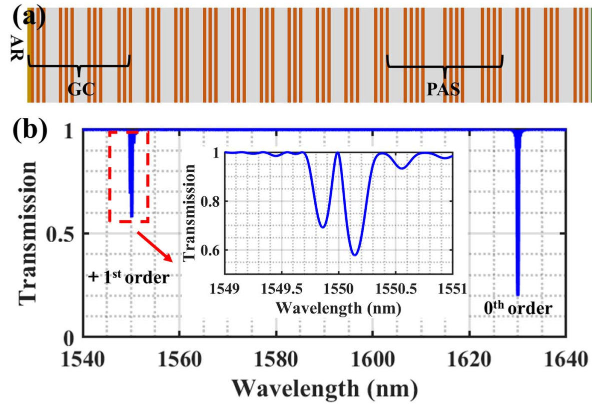

Fig. 2. (a) Schematic of the sampled grating with the ACPM structure implemented via the REC technique; (b) transmission spectra of the +1st-order subgrating calculated using the TMM with the length of the grating of 2500 μm. GR, grating compensation; PAS, phase-arranging section.

intended phase shift has been successfully synthesized via the equivalent CPM structure.

To suppress oscillation at the 0th-order subgrating, the seed grating is designed with a Bragg wavelength of 1630 nm, providing a large spectral detuning from the target operation band. The specific lasing wavelengths are defined by varying the sampling period from 3.551 to 5.448 μm. This configuration yields a 20-channel array covering a range from 1533 to 1563.4 nm with a uniform channel spacing of 200 GHz (1.6 nm). Figure 3 illustrates the schematic of the proposed laser.

Figure 4(a) presents a micrograph of the fabricated 20-channel high-power DFB laser array. The individual laser die features a cavity length of 2500 μm and a width of 250 μm, yielding a total array width of 4000 μm. The detailed structural schematic of a single DFB element is depicted in Fig. 4(b). To facilitate rigorous characterization of critical performance metrics—including fiber-coupled power, RIN, and linewidth—a representative chip was integrated into a standard 14-pin butterfly package. To ensure operational stability, the module incorporates a thermoelectric cooler (TEC) for precise temperature control and a dual-stage optical isolator to effectively suppress external optical feedback. High-efficiency optical coupling to the output fiber is realized through a precision lensing system. The internal

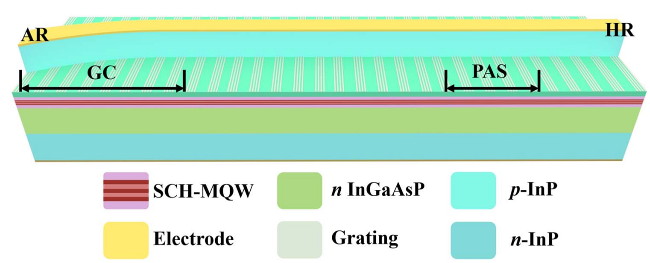

Fig. 3. Schematic of the proposed high-power DFB laser structure, featuring an asymmetric waveguide design and a 7° angled output waveguide. GC, grating compensation; PAS, phase-arranging section; SCH-MQW, separate confinement heterostructure multi-quantum well.

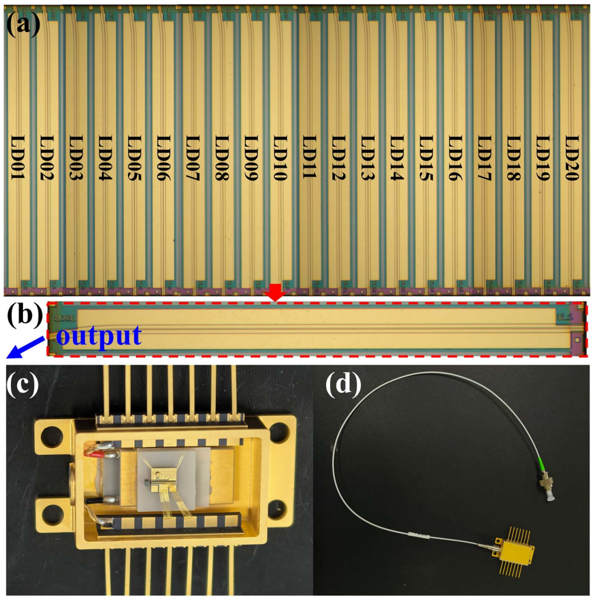

Fig. 4. (a) Microscopic top view of the proposed DFB laser array, (b) microscopic top view of one laser unit, (c) photograph of the internal structure of the butterfly package, and (d) photograph of the fully assembled butterfly module.

packaging configuration and the fully assembled module are shown in Figs. 4(c) and 4(d), respectively.

3. Device Characterization

To evaluate the fundamental static characteristics, the fabricated DFB laser array was die-bonded onto a C-mount heatsink using indium solder to ensure efficient thermal dissipation. The output emission was coupled into a lensed fiber and analyzed via an optical spectrum analyzer (Yokogawa AQ6370) with the submount temperature stabilized at 25°C. Figure 5(a) presents the superimposed lasing spectra of all 20 channels driven at an injection current of 500 mA. The array exhibits excellent uniformity, with every channel operating in a stable single longitudinal mode. As quantified in Fig. 5(b), the SMSRs are consistently high, exceeding 65 dB across the entire array. Furthermore, the lasing wavelengths were extracted and linearly fitted to assess the spacing accuracy, as depicted in Fig. 5(c). The resulting slope indicates an average channel spacing of 1.61 nm, which corresponds to a negligible discrepancy of only 0.01 nm from the design target. The maximum wavelength residual deviation from the linear fit is constrained to within 0.14 nm.

Subsequently, the output power performance was characterized under CW conditions with the C-mount temperature maintained at 25°C. A large-area optical power meter (Thorlabs PM100D) was utilized for accurate detection. Figure 6 illustrates the typical light–current–voltage (L–I–V) characteristics of a representative channel as the injection current is swept up to 1 A. The device exhibits a low series differential resistance of approximately 1.0 Ω. In terms of optical efficiency, the slope

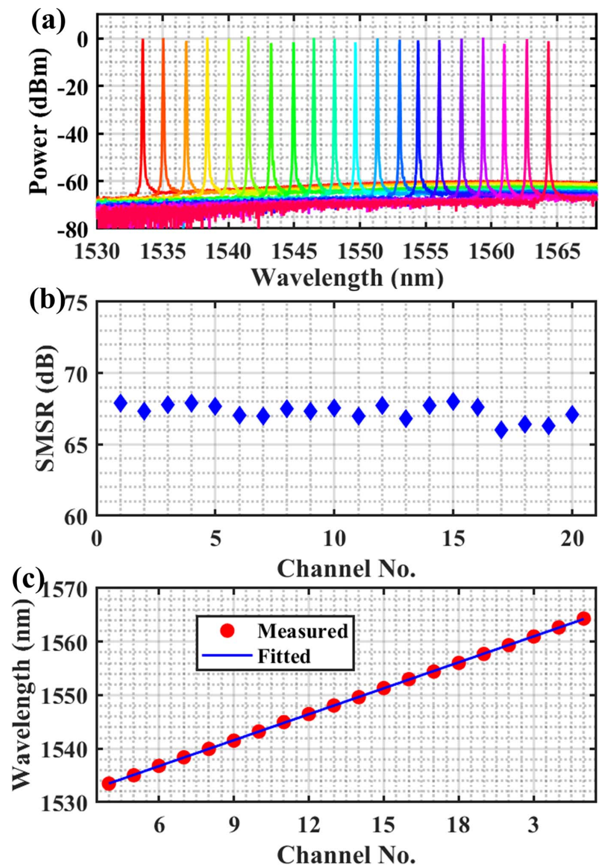

Fig. 5. (a) Superimposed optical spectra of all the 20 channels, (b) SMSRs of all the 20 channels, and (c) the measured lasing wavelengths and the fitted results.

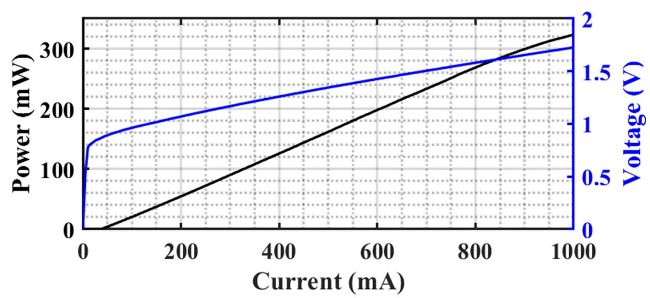

Fig. 6. Output powers and voltages of the selected typical channels when the bias currents are varied from 0 to 1 A.

efficiency is determined to be >0.4 W/A at an output level of 200 mW. Notably, the laser demonstrates robust high-power capability, delivering a maximum output power exceeding 320 mW at a bias current of 1 A.

Figure 7 depicts the measured far-field intensity distribution of the proposed DFB laser at a bias current of 500 mA. The FWHM divergence angles are 15.9° and 34.9° in the lateral and vertical directions, respectively. Such a confined far-field profile effectively reduces coupling losses, facilitating high-efficiency coupling into single-mode fibers.

To validate the practical utility of the device, a representative laser chip was integrated into a standard butterfly package, as previously depicted in Figs. 4(c) and 4(d). A comprehensive characterization series was subsequently performed on the packaged module, encompassing mode stability under varying drive currents and temperatures, fiber-coupled output power, RIN, and linewidth measurements.

We first examined the spectral stability as a function of injection current. Figure 8(a) illustrates the evolution of the lasing spectra as the drive current is swept from 100 mA to 1 A, with the case temperature tightly controlled at 25°C. A continuous red-shift in the lasing wavelength is observed, attributed to the internal heating effect. Crucially, the device maintains robust single-mode operation without any evidence of mode-hopping throughout the entire current range. Figure 8(b) quantifies this behavior, showing the wavelength shifting from 1559.23 to 1561.25 nm as the current increases. Based on this wavelength tuning characteristic, the thermal resistance from the active region to the cold side of the TEC is estimated to be approximately 15.3 K/W.

We further characterized the spectral and output power performance of the packaged module under varying operating temperatures. Figure 9(a) depicts the evolution of the lasing spectra at a constant injection current of 1 A across a temperature range from 25°C to 75°C. The lasing wavelength red-shifts with a measured temperature coefficient of approximately 0.1 nm/°C, while the SMSR is maintained above 65 dB throughout the entire temperature range, demonstrating excellent spectral purity.

Figure 9(b) summarizes the fiber-coupled output power characteristics from 25°C to 75°C. At room temperature (25°C), the output power exceeds 190 mW at 1 A. Remarkably, even at an elevated case temperature of 75°C, the device sustains an output power of greater than 150 mW at the same drive current. Additionally, the threshold current exhibits a moderate increase from 35 mA at 25°C to 85 mA at 75°C, consistent with the expected temperature dependence of the active region gain.

The RIN characteristics of the packaged device were subsequently evaluated using a high-precision noise measurement system (Sycatus). Measurements were conducted with the case temperature stabilized at 25°C over a bias current range of 200 to 500 mA. As shown in Fig. 10, the RIN level decreases as the injection current increases. Notably, when the drive current reaches 500 mA, the RIN is suppressed to a level below −160 dB/Hz. Such an ultra-low noise floor is highly advantageous for practical deployment in noise-sensitive high-speed optical interconnects.

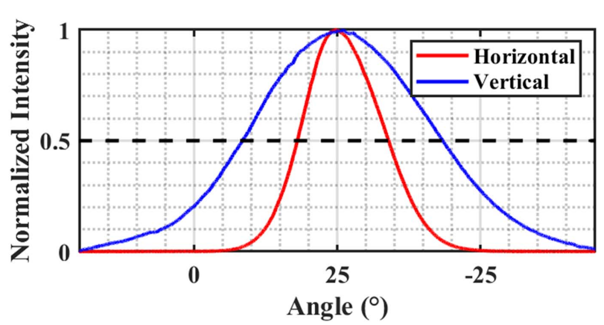

Fig. 7. Far-field intensity distributions of the proposed DFB laser biased at 500 mA.

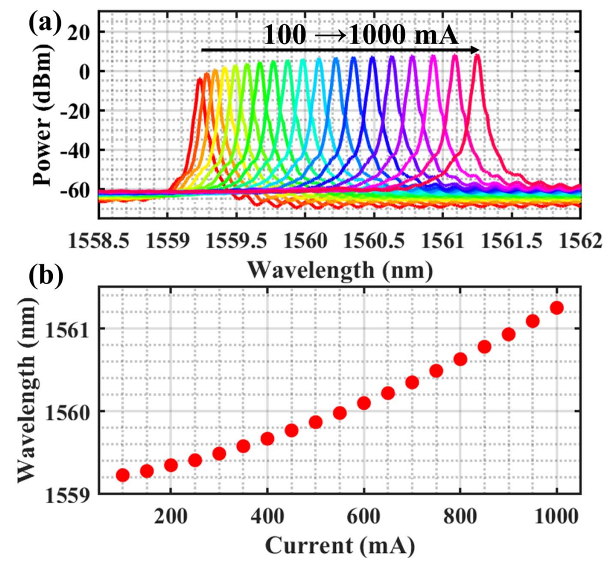

Fig. 8. (a) Lasing spectra measured at injection currents from 100 mA to 1 A at 25°C. (b) Peak wavelength versus current.

Finally, the spectral linewidth of the laser was characterized using the delayed self-heterodyne method. The experimental setup is illustrated in Fig. 11. The laser was powered by an ultra-low-noise current source (Newport LDX-3620B). The output from the packaged module first passes through an optical isolator to prevent back-reflections. The beam is then split into two paths: a delay arm consisting of a 25 km single-mode fiber to decorrelate the phase, and a frequency-shifting arm containing an acousto-optic modulator (AOM) operating at 80 MHz. The signals from both arms are recombined and detected by a high-speed photodetector, with the resulting beat signal analyzed by an electrical spectrum analyzer (ESA).

Figure 12(a) presents the measured beat note spectrum at a bias current of 300 mA and a temperature of 25°C. The spectrum exhibits a −20 dB bandwidth of 587 kHz. By fitting the beat note to a Lorentzian profile (where the intrinsic linewidth is $\Delta\nu = \Delta f_{-20\,\mathrm{dB}}/20$), the calculated intrinsic linewidth is approximately 29.35 kHz. Figure 12(b) summarizes the extracted Lorentzian linewidth over a drive current range of 200–500 mA. The linewidth remains consistently below 40 kHz across this range. To the best of our knowledge, this ultra-narrow linewidth combined with high output power represents one of the state-of-the-art results for DFB laser arrays. These characteristics make the device highly suitable for phase-sensitive applications, particularly in coherent optical communication systems and LIDAR systems. Compared to commercial state-of-the-art benchmarks like the FITEL FRL15TCW series (16-element array, 120 channels), our 20-channel array offers superior individual performance. Although the FITEL module delivers a maximum power of 17 dBm and a linewidth of 150 kHz, our

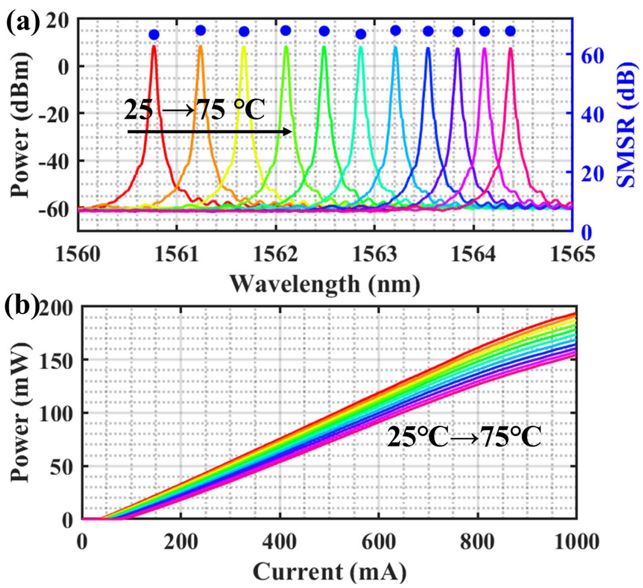

Fig. 9. (a) Optical spectra at the bias currents of 1 A with operating temperatures varied from 25°C to 75°C, and (b) output power when the bias currents are varied from 0 to 1000 mA when the operating temperature is varied from 25°C to 75°C.

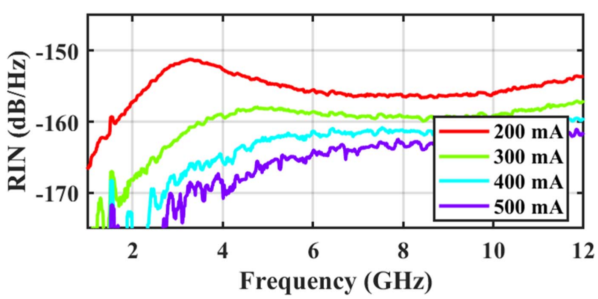

Fig. 10. Relative intensity noise at 200, 300, 400, and 500 mA, respectively.

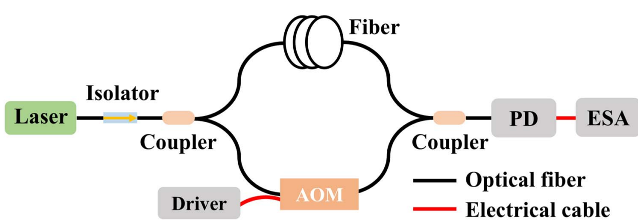

Fig. 11. Schematic of the delayed self-heterodyne linewidth measurement setup.

device achieves a significantly higher chip power of 25 dBm (>320 mW) and a narrower linewidth below 40 kHz (min. 29 kHz). Although currently utilizing separate outputs, these results provide a more robust high-power and high-purity foundation for next-generation optical I/O and coherent systems.

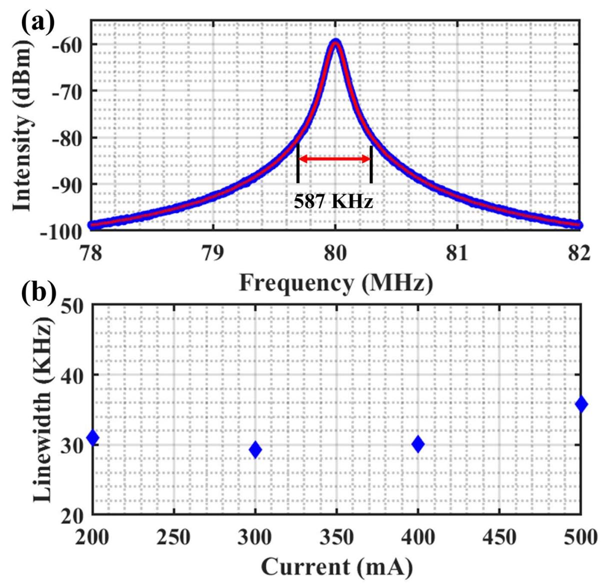

Fig. 12. (a) Measured beat spectrum at 300 mA (25°C) showing a 20 dB bandwidth of 587 kHz. (b) Calculated Lorentzian linewidth versus injection current.

4. Conclusion

We have experimentally demonstrated a high-power 20-channel DFB laser array operating in the C-band. To ensure robust single-mode stability, the device incorporates an equivalent ACPM grating based on the REC technique, which effectively mitigates the L-SHB effect. Additionally, a lightly n-doped passive waveguide is introduced to suppress free-carrier absorption losses, significantly enhancing slope efficiency. The laser array exhibits precise 200 GHz channel spacing with a wavelength deviation within 0.14 nm. It delivers superior power performance, achieving a fiber-coupled output power of >190 mW at 25°C and sustaining >150 mW at 75°C under a 1 A bias current. Furthermore, the array demonstrates excellent spectral purity with an intrinsic linewidth of 29 kHz and an RIN floor below −160 dB/Hz. These characteristics confirm the array's suitability for demanding applications in optical I/O, LiDAR, and free-space communications.

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